

Hand-Eye Calibration of a Laser Profile Scanner in Robotic Welding

Jindřich Liška
*Department of Cybernetics
University of West Bohemia
Pilsen, Czech republic
jinliska@ntis.zcu.cz*

Ondřej Vaníček
*Department of Cybernetics
University of West Bohemia
Pilsen, Czech republic
ovanicek@ntis.zcu.cz*

Michal Chaluš
*Department of Cybernetics
University of West Bohemia
Pilsen, Czech republic
chalus@ntis.zcu.cz*

Abstract—This paper presents a new version of the Hand-eye calibration method for a 2D laser profile scanner. To begin, the article brings a brief introduction to the problem and discusses the current state-of-the-art. Next, there is a description of the algorithm using a standard sphere as the calibration object and performing the computation of rotational and translational parts of the calibration transformation separately. The proposed algorithm is verified by multiple simulations to prove the compliance of the accuracy requirements and a real world application is presented to illustrate the benefit of the algorithm.

Index Terms—hand-eye calibration, laser profile scanner, robotics, welding

I. INTRODUCTION

Industrial robots find application in many sectors these days and they are no longer affordable just by large factories with mass production. They spread into small and medium-sized businesses, which however have their own specific requirements. Unlike mass producers, these smaller manufacturers require a compact and modular robotic workplace, which is able to switch between multiple different tasks with various tools in a short time [1].

This requirement is particularly important in the area of small-series robotic welding, in which this work was developed. It is stated that in the task of robotic welding of a single workpiece, the total time of preparation of the robotic system including tool changes, calibration, workpiece registration and trajectory learning occupies up to ninety percent of the total time spent on the workpiece [2]. This approach is therefore not very efficient and the machine time could be used in much more efficient way with the application of automation and advanced control mechanisms in a complex control system.

Sensors are an irreplaceable component of every automatic system and in order to use their measurements properly, they need to be calibrated. The calibration of the sensor itself, i.e. finding a relationship between physical measurements and the desired output, is usually performed by the sensors' producers, but the position of the sensor relative to the robot

flange needs to be evaluated by the control system. The controllers of the industrial robots usually include standard three-point methods for *tool center point* (TCP) calibration, but these methods are not applicable on most optical sensors because their TCP is located inside the sensor's case in the focus of the optical system and cannot be positioned to a reference point (e.g. towards a spike) in the workspace with various orientations. The specific requirements of the calibration of optical sensors are integrated in procedures that are referred to as *Hand-eye calibration* [3] - [7]. The main difference between standard calibration methods and hand-eye calibration algorithms is the necessity of a special calibration object with known geometry to calculate the position of a sensor above the object based on its measurements.

This paper presents a well-tried hand-eye calibration method which was developed as a component of a complex control system for robotic laser welding which due to the absence of a filling material requires high beam positioning accuracy with tolerance to errors of 0.1 mm at most.

II. BACKGROUND

This section delivers a brief summary of mathematical instruments which are necessary for operations with position in robotics and defines notation and emplacement of the coordinate systems used in the rest of this paper.

A. Homogeneous transformation matrix

The homogeneous transformation matrix is probably the most popular instrument for positional computations in robotics. Its main advantage over other instruments is the unification of both parts of motion, i.e. translation and rotation, into one mathematical operation - matrix multiplication.

The homogeneous transformation matrix consists of four segments, which define rotation, translation, projection and scale. In case of a standard three-dimensional space, the matrix is of dimensions $[4 \times 4]$ with the segments noted as in eq. (1). The indices 1 and 2 refer to the target *coordinate systems* (CS) where the upper index (1 in eq. (1)) notes the reference CS and the lower index (2 in eq. (1)) stands for the destination CS.

$$T_2^1 = \left[\begin{array}{ccc|c} R_2^1 & & & t_2^1 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad (1)$$

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The rotational part is defined by the upper left submatrix R_2^1 of dimensions $[3 \times 3]$ which is a standard orthogonal rotation matrix whose columns represent direction vectors of axes of the rotated CS2 in coordinates of the reference CS1. Translation is defined by the translational vector t_2^1 which is located in the first three elements of the fourth column and represents the coordinates of the origin of the destination CS2 in the reference CS1. The first three coefficients of the last row define perspective and the element on the position (4,4) changes the scale. The last row remains constant for purposes of robotics as shown in eq. (1) but it varies in other cases, such as computer graphics.

The homogeneous transformation matrix is a very compact tool for calculations with position. This applies also to evaluating inverse relation between coordinate systems. To evaluate the position and orientation of the reference CS in coordinates of the transformed CS based on a known transformation matrix T_2^1 , one simply calculates an inverse matrix as shown in eq. (2). Furthermore, this operation can be made computationally simpler using the orthonormality of a rotation matrix which implies equivalence of inversion and transposition of the matrix.

$$T_1^2 = (T_2^1)^{-1} = \left[\begin{array}{ccc|c} (R_2^1)^T & & & -(R_2^1)^T \cdot t_2^1 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad (2)$$

Another useful property of the homogeneous transformation matrix is the possibility of concatenating transformations as shown in eq. (3).

$$T_3^1 = T_2^1 \cdot T_3^2 \quad (3)$$

This property is favourably utilized for example in geometrical modelling of industrial robots where the end effector's position T_e^B is calculated based on known transformations between particular joints and the calibration transformation expressing the end effector's position on the robot flange. While the position of the robot flange T_N^B is calculated by the robot controller, the last factor T_e^N needs to be calculated by the calibration algorithms (hand-eye calibration in case of optical sensors).

B. Coordinate systems definition

In order to understand the steps in the next sections properly, it is necessary to present the relationship between significant coordinate systems. Figure 1 shows the placement of important systems. The coordinate systems are denoted with capital letters in italics, where *B* stands for base (*world*) CS, *N* represents the robot's flange CS, *S* represents sensor CS and *E* represents the CS of the calibration object. Figure 1 contains an industrial robot equipped with a laser welding head carrying a laser profile scanner and a standard metal sphere which was used as a calibration object.

The origin of CS *B* is located on the axis of rotation of the first joint inside the robot's base, the *z* axis of CS *B* is concurrent with the rotation axis of the first joint and the *x* axis of CS *B* is aimed so that the origin of CS *N* (robot flange) lies in (*xz*) plane of CS *B* when the joint coordinate of the first joint is zero ($J1 = 0$). The origin of CS *N* is located

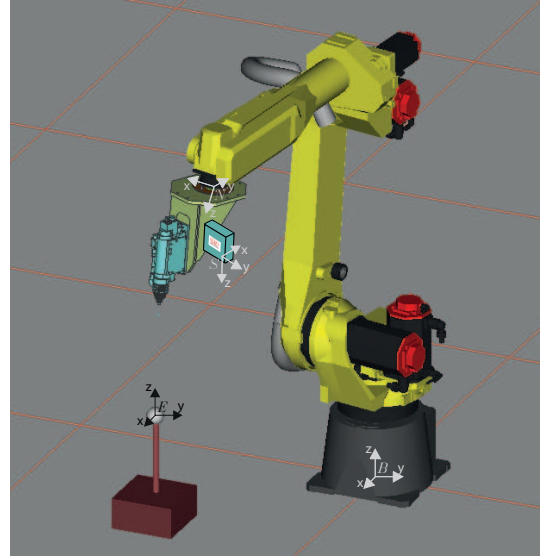


Fig. 1. Significant coordinate systems

in the middle of the robot's flange with *z* axis aiming out of the flange plane. The origin of CS *S* is located in the focus of the optical system of the laser profile scanner with *z* axis aiming out of the camera and *x* axis perpendicular to *z* in the measuring plane (the *y* coordinate of the 2D scanner output is always zero). The origin of CS *E* is located in the center of the calibration sphere with the same orientation as the base coordinate system.

III. CALIBRATION

The process of evaluating the calibration transformation T_S^N represents a search for the relative position of the robot's flange and the sensor's focal point in three-dimensional space which generally has six degrees of freedom. Mathematically, this process can be summarized as solving a matrix equation $AX = XB$ where *A* and *B* represent a relative motion of the robot's flange and the sensor respectively and *X* stands for the sought calibration transformation. The idea of the hand-eye calibration equation is illustrated by Figure 2. The calibration matrix consists of a rotational and a translational part as shown in eq. (4).

$$X = T_S^N = \left[\begin{array}{ccc|c} R_S^N & & & t_S^N \\ 0 & 0 & 0 & 1 \end{array} \right] \quad (4)$$

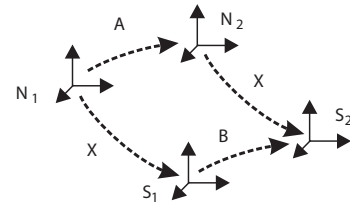


Fig. 2. Hand-eye calibration illustration

The calibration matrix equation has six independent parameters and it has been shown in [3] that it is necessary

to gather at least three measurements with different robot poses to find a unique solution to this equation. However, because of the presence of measurement noise, it is advisable to capture more positions and calculate the calibration transformation based on optimization of an error criterion.

In order to calculate the relative motion of the sensor (or tool in general), it is necessary to use some reference calibration object which allows to either ensure that the TCP is in the same position for all the measurements (e.g. a spike used with standard algorithms which is, however, inapplicable on optical sensors) or calculate the position of the sensor relative to the object based on the sensor's measurements. Several cuboidal calibration objects have been developed [5], [6], [7] but all of them bring additional difficulties regarding solution of non-linear systems of equations and there is a real risk that even if the measured data is correct, the calculated relative position of the sensor above the object is wrong [8]. This is why we decided to use a standard sphere as the calibration object as it requires to solve *only* a quadratic equation and to choose one out of two possible solutions. A sphere was used as the calibration object also in [10] while the system configuration was different and the approach did not consider the possible iterative improvement.

A. Sphere center localization

Evaluation of the sphere center in the coordinate system of the laser profile scanner from a single measured profile is an essential step of calculating the calibration transformation. The idea of the evaluation of the sphere center position is depicted by Figure 3. The red arc represents the data obtained from the laser profile scanner, the center of the calibration sphere is denoted as C_E , r_E stands for its a priori known radius, C_S is the center of the interpolated circle in the measurement plane and r_S represents the interpolated circle's radius. The interpolation process is based on the least squares best circle fitting algorithm presented in [9].

The algorithm calculates a 3D position from the 2D data based on the following geometrical axioms:

- 1) a non-empty (and non-punctual) intersection of a sphere and a plane always forms a circle
- 2) the normal vector of such plane is parallel to the directional vector of the line between the center of the sphere C_E and the center of the intersected circle C_S
- 3) the distance between the two centers can be calculated from the known sphere radius r_E using the Pythagorean theorem $h = \sqrt{r_E^2 - r_S^2}$.

Hence, the x and z coordinates of the sphere center C_E in the sensor CS can be obtained from a circular interpolation of the measured profile and the absolute value of the y coordinate can be calculated from the known radius of the calibration sphere and the evaluated radius of the fitted circle (5).

$${}^S C_E = ({}^S C_{Sx}, \pm h, {}^S C_{Sz}) \quad (5)$$

Thus, the only uncertainty is the sign of the y coordinate (i.e. whether the sphere center is in front of the measuring plane or behind it) and it needs to be decided based on some additional information.

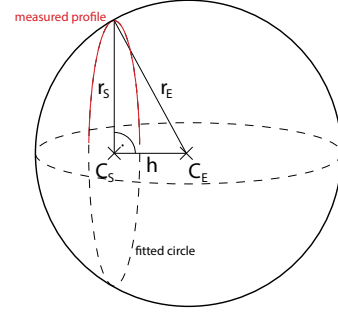


Fig. 3. Sphere center localization scheme

This additional information could be obtained from an additional sensor, e.g. a camera rigidly mounted next to the laser profile scanner aiming the same direction as the scanner, but it would increase the cost of the system and reduce its compactness, which was the reason why we decided to apply a more compact method. Our method uses a rough initial calibration, which is obtained using a modified three-point method and is performed manually, and a defined initial scan of the calibration sphere. After this, the position of the sphere center ${}^S C_E$ in the sensor CS can be chosen based on the distance from the initially obtained sphere center position in the base CS.

B. Rotational part

The rotational part of the calibration transformation is represented by the rotation matrix R_S^N in eq. (4) and needs to be calculated first as the estimation of the translational part assumes knowledge of the rotational part.

Unlike the cuboidal calibration objects used in [5], [6], [7], the sphere doesn't provide any information about relative orientation. However, it is possible to gather data from defined motions over the calibration sphere which allow us to calculate the rotation matrix R_S^N as shown in [10].

Assuming the robot moves the origin of CS N (flange) linearly along a given direction defined by a unit vector ${}^N v$, the calculated sphere centers should also be distributed on a line defined by a directional vector ${}^S v$ in the sensor's CS S . According to the properties of rotation matrices, these vectors are related as

$${}^N v = R_S^N \cdot ({}^S v). \quad (6)$$

Generally, a vector is defined by two points, but in order to reduce the influence of the measurement noise, we used multiple measurements and calculated the vectors using the least squares line fitting.

By moving the robot flange linearly in three different (ideally orthogonal [10]) directions, the equation (6) can be rewritten as (7)

$$[{}^N v_1 \ {}^N v_2 \ {}^N v_3] = R_S^N \cdot [{}^S v_1 \ {}^S v_2 \ {}^S v_3] \quad (7)$$

and finally applying the substitutions ${}^N V = [{}^N v_1 \ {}^N v_2 \ {}^N v_3]$ and ${}^S V = [{}^S v_1 \ {}^S v_2 \ {}^S v_3]$ the rotation matrix can be calculated according to eq. (8).

$$R_S^N = {}^N V \cdot ({}^S V)^{-1} \quad (8)$$

Due to the presence of measurement noise, the calculated matrix R_S^N might not be precisely orthogonal, which is adjusted by applying the Gram-Schmidt algorithm.

C. Translational part

Unlike the solution of the rotational part, which is solved differently in each work [3], [5], [6], the translational part of the calibration transformation is sought generally the same way as a minimum of a square criterion, which is defined by eq. (9). The minimal argument t_S^N of this criterion minimizes the euclidean distance of the calibration equation $AX = XB$ based on $n + 1$ measured positions of the robot flange and the reference point position in sensor CS assuming the knowledge of the rotation matrix R_S^N .

$$\eta(t_S^N) = \sum_{i=1}^n \|(R_{N_{i+1}}^N - I_{[3 \times 3]}) \cdot t_S^N - R_S^N \cdot t_{S_{i+1}}^N + t_{N_{i+1}}^N\|^2 \quad (9)$$

Minimizing this criterion leads on a standard least squares solution

$$t_S^N = (C^T C)^{-1} C^T b, \quad (10)$$

where the regression matrices are composed as follows.

$$C = \begin{bmatrix} R_{N_1}^N - I_{[3 \times 3]} \\ R_{N_2}^N - I_{[3 \times 3]} \\ R_{N_3}^N - I_{[3 \times 3]} \\ \vdots \\ R_{N_{n+1}}^N - I_{[3 \times 3]} \end{bmatrix}, \quad b = \begin{bmatrix} R_S^N \cdot t_{S_1}^N - t_{N_1}^N \\ R_S^N \cdot t_{S_2}^N - t_{N_2}^N \\ R_S^N \cdot t_{S_3}^N - t_{N_3}^N \\ \vdots \\ R_S^N \cdot t_{S_{n+1}}^N - t_{N_{n+1}}^N \end{bmatrix} \quad (11)$$

By combining the results of equations (8) and (10) it is now possible to compose the sought calibration transformation matrix according to eq. (4).

IV. ALGORITHM

Our proposed algorithm consists of the following steps:

- 1) Set the initial (*manual*) calibration $T_{S_{init}}^N$ - e.g. based on a CAD model of the tooling with the sensor or by performing a rough three-point method based manual calibration using the native function of the robot controller.
- 2) Perform an initial scan by positioning the sensor linearly above the calibration sphere, find the maximum radius and save the position of the greatest radius' center C_{E0} .
- 3) Place the sensor vertically above the sphere, measure $3 \times N_{rot}$ positions during positioning the sensor in the direction of the axes of the base CS, calculate vectors ${}^N v_i$ and ${}^S v_i$ and find the rotational part of the calibration according to eq. (8).
- 4) Place the sensor into N_{trans} randomly¹ generated positions, calculate $T_{N_{n+i}}^N$, $T_{S_{n+i}}^N$ and evaluate the translational part of the calibration transformation T_S^N according to eq. (10).
- 5) Compose the calibration transformation matrix according to eq. (4), compare it to the initial calibration $T_{S_{init}}^N$

¹it is useful to choose only those positions which ensure non-empty intersection of the laser plane and the sphere located at C_{E0}

- if the convergence requirements are met, finish the algorithm, otherwise perform all calculations from step 2) with $T_{S_{init}}^N = T_S^N$.

V. SIMULATIONS

This section is dedicated to the presentation of the results of the simulations performed prior to the practical use of the algorithm on a real industrial system. These simulations considered realistic inaccuracies in the robot's positioning, laser sensor measurement noise and a wrong initial (*manual*) calibration.

A. Single simulation

The first part of this section is devoted to the presentation of the results of a single run of our simulation.

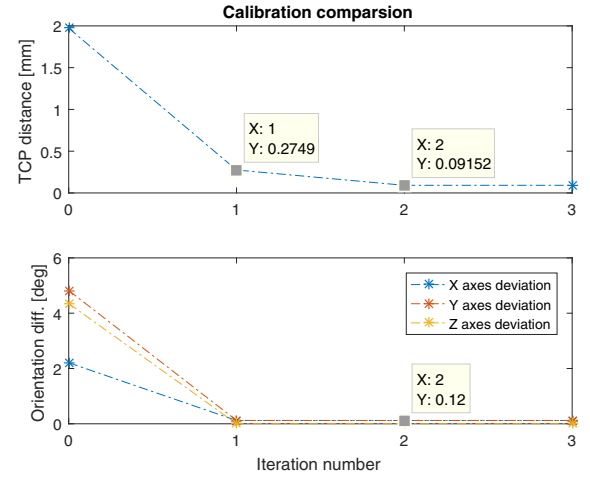


Fig. 4. Comparison of calibration transformations

The initial parameters of the simulation are summarized in Table I. This set-up corresponds to reality while considering a minimal accuracy of the manual calibration (in praxis the manual calibration is usually more accurate - differences of up to 1 mm resp. 1°). The simulated measurement noise of the sensor corresponds to the real noise of the measured profiles (experimentally established at $\sigma_l = 0.16$ mm which corresponds to the precision declared by the manufacturer) and the inaccuracy of the robot was set based on the declared repeatability of the robot of $\sigma_r = 0.08$ mm.

TABLE I

Initial simulation parameters		
Parameter description	Real value	Set initial value
Translational part (XYZ) [mm]	[2, 74, 16]	[3.5, 73, 16.8]
Rotational part (WPR) [deg]	[128, 69, -97]	[130, 67, -99.5]

The resulting accuracy presented by Figure 4 was achieved by simulating $N_{rot} = 20$ profiles for each axis during orientation calibration and $N_{trans} = 300$ profiles for calibrating the translational part. This figure shows that the calibration process was finished after three iterations of the proposed algorithm as the evaluated calibration transformation from the second iteration was identical to the third one. The algorithm

produced resulting TCP inaccuracy of $d_{trans} = 0.09 \text{ mm}$ and a rotational error of about $\delta_{axes} = 0.1^\circ$ for each axis. It is important to note that the presented rotational error is computed as the deviation of the directional vectors of the axes of the evaluated CS S from the directional vectors of the axes of the correct CS S as shown in eq. (12) which is more predicative than a simple subtraction of the WPR values.

$$\delta_{axes} = \arccos \left(\frac{|\vec{s}_{real} \cdot \vec{s}_{eval}|}{|\vec{s}_{real}| |\vec{s}_{eval}|} \right) \quad (12)$$

B. Monte Carlo simulations

The second part of this section presents the results of Monte Carlo simulations of the proposed algorithm which were performed in order to establish the dependence between the number of profiles N_{trans} used in the algorithm and the accuracy of the resulting calibration transformation.

These simulations prove that the accuracy of the calculated calibration transformation depends on the number of profiles used for calculating the translational part. The simulation was executed for $N_{trans} = \{50, 100, 250, 500, 1000, 2000\}$ profiles and the algorithm was executed $N_{sim} = 100$ times for each selection of N_{trans} . The precision histogram shown in Figure 5 illustrates that the accuracy grows with the increasing number of profiles. However, this dependence does not seem to be direct.

The presented simulation showed that the noise present in the system may cause an inaccuracy in the calculated calibration of more than 0.6 mm if the number of profiles is $N_{trans} \leq 250$. The selection $N_{trans} = 500$ already shows good precision and produces the average $\bar{t}_{error} = 0.1176 \text{ mm}$ and $\max(t_{error}) = 0.335 \text{ mm}$ while selecting $N_{trans} = 1000$ or even $N_{trans} = 2000$ does not bring any significant refinement producing $\bar{t}_{error} = 0.1016 \text{ mm}$ and $\max(t_{error}) = 0.2726 \text{ mm}$. The statistical details of this simulation are summarized in Table II

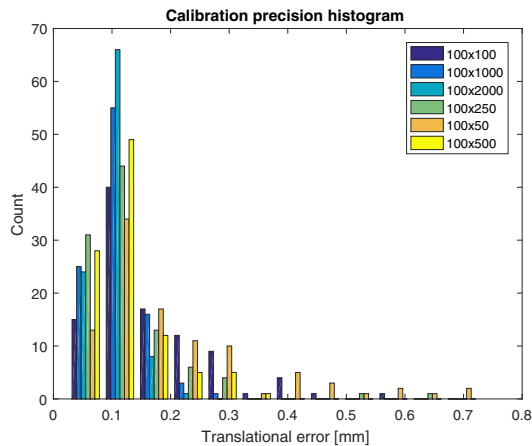


Fig. 5. Calibration precision histogram

These simulations showed that a reasonable number of profiles used to calculate the translational part with the desired accuracy of 0.1 mm is around five hundred. We executed an analogical simulation in order to determine the

TABLE II

Statistical parameters			
N_{trans}	$\text{mean}(t_{error})$ [mm]	$\text{std}(t_{error})$ [mm]	
50	0.2019	0.15	
100	0.1642	0.0988	
250	0.1275	0.0853	
500	0.1176	0.0623	
1000	0.1113	0.0443	
2000	0.1016	0.0286	

number of profiles needed for the rotational part and it showed that the reasonable value is $N_{rot} = 25$.

VI. REAL USE

As mentioned above, we used a standard sphere as the calibration object, namely a steel bearing ball with a diameter of 55 mm and a precision grade of G40 which implies sphericity and ball diameter variation of $1 \mu\text{m}$ at most. Additionally, the surface of the sphere was covered with a thin layer of a white matte spray in order to reduce measurement noise of the laser profile scanner. The MetraLight TLE1 profile scanner was mounted on a laser welding head carried by the FANUC M20iA/20M industrial robot. A photo of the whole system is shown in Figure 6.

The whole system was controlled by our control software for intelligent robotic welding which unifies the control of the robot with data acquisition and processing. After manually obtaining the initial calibration transformation, the system followed the steps presented in Section IV with the presumably ideal parameters $N_{rot} = 25$, $N_{trans} = 500$ and calculated the sought calibration transformation T_S^N .

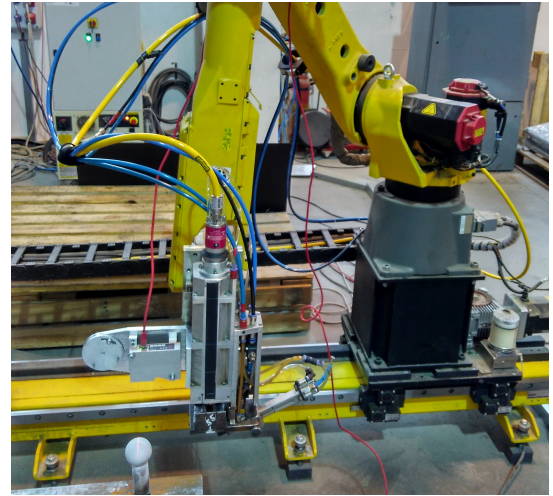


Fig. 6. Global view of the system

The system was afterwards used for scanning a turbine blade intended for laser quenching in order to create a 3D model for definition of the quenching areas. The robot was following pre-defined line segments along different sides of the blade and obtaining profile measurements of the surface in order to gather spatial data which could be formed into a point cloud and used for creating the 3D model of the

scanned blade. The role of the calibration transformation is crucial in such operations. Each profile measured by the sensor is converted into the world coordinate system using the calibration transformation and in case of its inaccuracies, the produced model would be very inaccurate (if not unusable).

The benefit of the calibration algorithm is illustrated by Figure 7 which compares the alignment of four planar scans in the area of the leading edge of the blade. This comparison proves that even a small inaccuracy in the calibration transformation can cause significant inconsistency when applied to data in larger volumes.

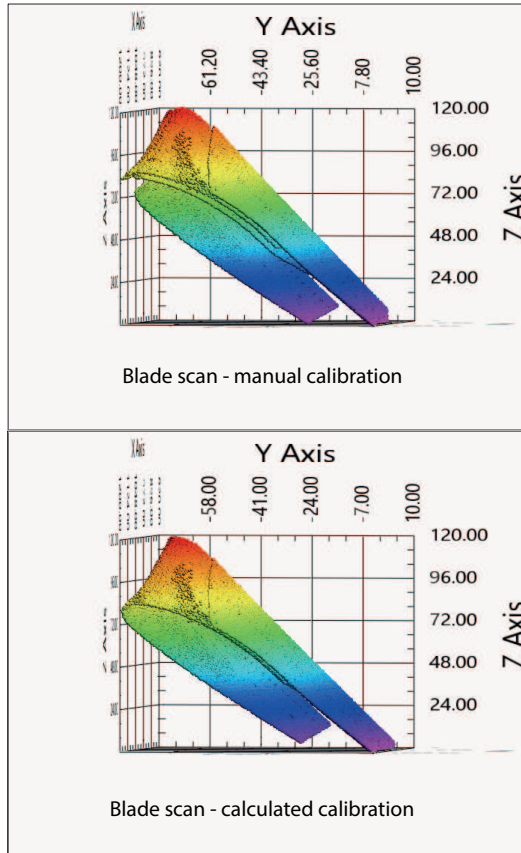


Fig. 7. Blade scan: manual and calculated calibration comparison

VII. COMPARISON TO RELATED METHODS

Related hand-eye calibration methods for profile scanners include methods using cuboidal calibration objects [5], [6], [7] and standard spheres [10]. Although the cuboidal objects provide the full 6 DoF information about the sensor position, they also bring the necessity of numerical solution of non-linear systems of equations which may cause additional inaccuracies. Computationally, these methods perform calculation of the rotational part from the same measurements as for the translational part, which makes the data acquisition simpler. On the other hand, the robot must avoid sensor poses implying ambiguity of the numerical solution which makes the position planning notably more complicated [8]. The sphere-using methods, including our method, avoid these complications by using a standard sphere as the calibration

object and devote more effort to calculate the rotational part of the calibration. The challenging part of the sphere center localization is the choice of the third coordinate in the sensor CS. Authors of [10] apparently used a rough knowledge of the sphere position on the robot flange which we had to substitute by initial scan and iterative calculations due to more complex configuration of our system.

In terms of accuracy, i.e. the deviation of the calculated calibration from the correct solution, our method is comparable to the most accurate approaches according to the results presented in publications cited above. The accuracy of the calibration however depends on the accuracy of the calibrated sensor which may vary between particular authors.

VIII. CONCLUSION

We have presented a new iterative version of the Hand-eye calibration method for a 2D laser profile scanner. Our approach uses a standard sphere as the calibration object and the algorithm performs defined steps to calculate the rotational and the translational part of the calibration transformation separately. The simulations, considering a real intensity of measurement noise and inaccuracies, showed an accuracy of 0.1 mm and 0.1 degree on average and the example of the real application also proved good precision of proposed algorithms, although the real accuracy cannot be quantified due to the unavailability of the real correct calibration.

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